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Efficiency analysis of engineering classes: A DEA approach encompassing active learning and expositive classes towards quality education

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ABSTRACT

The science, technology, engineering, and mathematics (STEM) education research delves into the core of sustainable development goals (SDGs), including the pillars of quality education (SDG4), robust economic growth (SDG8), and diminished inequalities (SDG10). These pursuits stand as keystones in sculpting inclusive societies and bridging societal gaps. While previous studies utilising data envelopment analysis (DEA) have explored educational performance mainly from a macro-perspective, there is a lack of micro-perspective investigation. Our study aims to fill this gap by proposing a DEA approach to assess the relative efficiency of engineering classes. We analysed 70 classes covering 38 subjects in the first semester of 2022 at a South American school. Methodologically, we employed the slack-based measure (SBM) model under the benefit of doubt (BoD) condition. Unlike prior research, we analysed classes' relative performance considering different pedagogical approaches - 11 active-learning classes (15.7 %) and 59 passive-learning classes (84.3 %). Our results showed that 18 classes were efficient (25.7 %). Active classes were more efficient, but few subjects maintained similar efficiencies for all classes. Moreover, efficient classes were concentrated in the last two years prior to graduation (57.9 %). This may represent an additional barrier for low-income students, who tend to drop out in the first years. The findings support several improvement recommendations, such as integrating digital technologies, boosting active learning opportunities, and bolstering classes in foundational subjects. Also, implications for researchers, decision- and policy-makers are discussed. Our approach can be replicated in diverse educational contexts, enabling the identification of strengths and weaknesses for more efficient educational management.

1. Introduction

The 17 Sustainable Development Goals (SDGs) serve as a global roadmap for tackling urgent social, economic, and environmental challenges, emphasising the interconnectedness of issues and the necessity for international collaboration to achieve a fairer, more prosperous, and sustainable future (United Nations, 2023). Focusing science, technology, engineering, and mathematics (STEM) education research on SDGs, like quality education (SDG 4), economic growth (SDG 8), and reduced inequalities (SDG 10), promotes inclusive societies, enabling everyone to thrive and participate in shaping a better world. Research in STEM education is pivotal for advancing these SDGs by ensuring equal access to quality education, empowering individuals from diverse backgrounds to contribute meaningfully to sustainable development and bridging gaps between privileged and underprivileged communities (De La Hoz et al., 2021).

Data envelopment analysis (DEA) is firmly established as a methodological tool for evaluating education performance. For instance, Cossani et al. (2022), Coupet (2017), Egorov and Serebrennikov (2023), Szuwarzyński (2019), and Taylor and Harris (2004) examined higher education performance across different countries. Other studies, like Barahona-Urbina and Barahona-Droguett (2023), Köksal and Nalçaci (2006), and Lee and Johnes (2022), focused on departmental efficiency within higher education institutions. However, there remains a gap in studies that explore micro perspectives, such as the efficiency of specific subjects within courses or individual classes within university subjects.

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An exception is Montoneri et al. (2012), who utilised a DEA CCR (an acronym of the authors Charnes, Cooper, and Rhodes, 1978) model to assess English classes at a Taiwanese university, identifying key performance indicators to enhance instructor performance and inform educational policy development. A point to highlight is that the CCR model considers proportional changes among measures. Micro-perspective research is crucial as it informs operational, tactical, and strategic educational decisions. Notably, no prior study has investigated the relative performance of engineering classes.

Efficiency can be evaluated using several methods, from traditional statistical analyses to complex DEA models (Lampe and Hilgers, 2015). We selected the slack-based measure (SBM) model by Tone (2001) under the variable returns to scale (VRS) assumption and benefit of doubt (BoD) condition (with unitary input) to create a composite index and evaluate educational efficiency. Compared to other efficiency evaluation methods (such as stochastic frontier analysis—SFA), the DEA models are non-parametric and handle multiple outputs and inputs. This means it is unnecessary to assume a specific functional form of a production frontier, enabling efficiency analysis in cases where the functional form of the technology is unknown or complex (Lampe and Hilgers, 2015). In addition, the SBM does not consider proportional improvements of inputs or outputs to achieve a goal based on the benchmarking and it accounts for slacks directly, offering more precise insights into specific areas of inefficiency and enabling targeted improvements and actionable insights for educational decision-makers. So, it offers distinct advantages when evaluating different multicriteria aspects, such as teaching quality and student outcomes, because improvements of these criteria may not be uniform across all pedagogical aspects.

In this context, our paper aims to propose and implement a straightforward approach for assessing the relative efficiency of higher education classes using DEA. Methodologically, we utilise indicators based on Montoneri et al. (2012) and employ the SBM-VRS-BoD model to construct a composite index. A BoD model has been previously utilised in an educational macro-perspective by Szuwarzyński (2019) to examine the relative performance of public Australian universities. However, this approach by Szuwarzyński (2019) has not been employed in a micro-perspective investigation before, i.e. there is a lack of studies evaluating classes instead of universities, highlighting the novelty of our approach.

The proposed DEA index (SBM-VRS-BoD) represents a pioneering approach in this domain, where the main novelty relies on assessing the relative efficiency of engineering classes while encompassing both active learning and passive pedagogical methods. It integrates dimensions of teaching and learning processes and inclusion toward inclusive quality education (SDG4) in a relative efficiency composite index, without assuming proportional changes among measures, enabling a comprehensive evaluation of class efficiency. We also evaluate the contribution of these several measures to the efficiency of classes. This multicriteria approach offers insights into student performance, pedagogical strategies, and the integration of digital technologies in learning. In our study, BoD operates under the principle of "the higher, the better," treating all measures as outputs and employing only unitary input to create a composite index, thus avoiding assumptions regarding the production perspective. This feature can be advantageous, as it evaluates efficiency without explaining how inputs generate outputs within the SBM-VRS-BoD framework.

Regarding the application, Montoneri et al. (2012) exclusively analysed passive-learning classes using a traditional DEA model, where students receive information primarily through lectures and direct instruction (i.e., expositive classes). In contrast, we employed an advanced DEA model to assess a sample of 70 classes encompassing 38 engineering subjects at a northern Chilean engineering school, comprising all subjects and classes for the first semester 2022. Among these, 11 were active, and 59 were passive, representing 15.7 % and 84.3 %, respectively. This study marks the first instance of benchmarking classes with differing pedagogical approaches using DEA.

Our application is pertinent as Chile boasts one of Latin America's highest gross enrolment rates in higher education (Martin and Furiv, 2021). However, its educational system is undergoing a comprehensive and contentious reform aimed at broadening access to low-income students, potentially addressing the country's significant challenge of social inequality (Espinoza et al., 2022; Mendoza, 2020). The proposed methodology can be replicated in countries grappling with similar issues of inequality and quality in higher education, particularly those in the developing world.

Moreover, our findings affect practitioners, higher education policymakers, and researchers. For instance, our research highlights the inefficiency of passive classes in the first year. This supports recommendations for improving the educational context, such as elevating the qualifications of teaching staff, particularly for first-year instruction, reassessing class size and evaluation methods, better integrating digital technologies, promoting active learning opportunities, and strengthening foundational subjects like programming, statistics, and thermodynamics.

This paper unfolds across five compelling sections. Firstly, a concise literature review and contextualisation set the stage. Then, the methodology is meticulously outlined in the third section. The fourth section unveils the results and sparks engaging discussions. Finally, the fifth section encapsulates conclusions, unveils limitations, and ignites pathways for future research.

2. Background

2.1. Sustainable development goals (SDGs)

The SDG 4 aims for inclusive and equitable education and promotes lifelong learning opportunities for all (United Nations, 2023). This SDG is divided into 11 targets. Among them, target 4.4 states, "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship" (United Nations, 2023), measured by the indicator of "Proportion of youth and adults with information and communications technology (ICT) skills, by type of skill" (United Nations, 2023). This target is one of the main focus of our paper, since we investigate the pedagogical performance of engineering classes in the Latin American context.

The SDG 8 promotes sustained, inclusive, sustainable economic growth, full and productive employment, and decent work (United Nations, 2023). SDG 8 is divided into 12 targets. Target 8.2 states, "achieve higher levels of economic productivity through diversification, technological upgrading and innovation, focusing on high-value-added and labour-intensive sectors" (United Nations, 2023). This target correlates with our research goal, since high-quality STEM education increases the number of qualified job offers, enabling the emergence and growth of high-value-added sectors (OECD, 2021).

Furthermore, active and passive pedagogical approaches are two different methods of teaching that may be applied in engineering education. While passive learning is a more traditional approach, where students receive information from the instructor through lectures and other forms of direct and expositive instruction, active learning is a student-centred approach, where students actively engage in the learning process, often through problem-solving, group work, and other interactive activities (Guerra and Mesa, 2017). Active learning increases the student's analytical thinking and creativity, and develops essential skills for entrepreneurship and labour-intensive activities (Graaff and Kolmos, 2017), prescribed by targets 4.4 and 8.2.

Finally, quality education (SDG 4) establishes the pathway for decent jobs and entrepreneurship (SDG 8), which reduces inequalities (SDG 10). The SDG 10 is committed to reducing inequalities within and among countries. This SDG is divided into 10 targets. Among them, two are related to our paper: 10.2, "By 2030, empower and promote the

social, economic and political inclusion of all, irrespective of age, sex, disability, race, ethnicity, origin, religion or economic or other status" (United Nations, 2023), and 10.4, "Adopt policies, especially fiscal, wage and social protection policies, and progressively achieve greater equality" (United Nations, 2023). Our study is also a timid initiative to support increasing the number of women taking STEM courses (4.2) because we did not neglect the percentage of women in classes. Besides, our study encompasses target 10.4 more, because the Chilean educational system has been undergoing a comprehensive reform intended to extend access to low-income students (Espinoza et al., 2022; Mendoza, 2020), which can be seen as a social protection policy. In this regard, our results could guide future long-term inclusive policies.

2.2. Data envelopment analysis (DEA) in higher education

The first data envelopment analysis (DEA) model was created in 1978, and called CCR - the acronym of the authors' names: Charnes, Cooper, and Rhodes (Charnes et al., 1978). The CCR model was based on Farrell's work and his definition of the efficiency frontier (Farrell, 1957) for measuring productive efficiency. To give an example of the application of the CCR model, one suggestion made by Charnes et al. (1978) was to analyse the efficiency of several schools, seen as decision-making units (DMUs). In DEA models, DMUs are associated with multiple measures, usually classified as "inputs" and "outputs". The DEA results are typically translated into an index between 0 and 1 for each DMU, where a DMU with 1 is 100 % efficient, and the DMUs with other results have a relative efficiency percentage. DEA models commonly require a choice of orientation, i.e., input minimisation or output maximisation (Cook et al., 2014). At a later date, (Banker et al., 1984) extended this model, proposing a new version (BCC) for the case of variable returns to scale (VRS) to estimate purely technical efficiency. It is worth noting that both traditional DEA models (CCR and BCC) were initially thought to be applied in education evaluation (Banker et al., 1984; Charnes et al., 1978).

Since the 1970s, several advanced DEA models have been developed, and different applications have been recorded in education assessment (Cook et al., 2014). Besides the choice of model and measure, these applications also vary in terms of education level and perspective. Relevant examples are presented and discussed in this subsection.

Considering the country-level perspective, Taylor and Harris (2004) applied traditional DEA models, experimenting with different orientation configurations, to measure the relative performance of 21 South African universities. Coupet (2017) applied traditional DEA models to investigate the impact of varying funding sources on North American Historically Black Colleges and Universities. Egorov and Serebrennikov (2023) proposed a stochastic DEA model for studying the effects of budget and the makeup of the student body and academic staff on the efficiency of Russian universities. The authors concluded that the three factors positively and significantly impact universities' performance. Szuwarzyński (2019) proposed the benefit of doubt (BoD) condition to measure the performance of 37 public Australian universities. The BoD condition is a DEA model (traditional or advanced) with unitary inputs (or outputs). Besides, once the BoD condition is applied to an SBM model, choosing an orientation is unnecessary – which can be seen as another advantage (Dyckhoff and Souren, 2022).

Going a step further toward the micro perspective, the internal performance of universities has been investigated. Barahona-Urbina and Barahona-Droguett (2023) applied traditional DEA models to explore the relative performance among departments at a public Chilean university. The results indicated that the engineering department was an efficiency benchmark. Köksal and Nalçaci (2006) applied a multi-criteria, traditional DEA models and sensitivity analysis to investigate the relative efficiency of departments at a Turkish university. The authors pointed out that one of the limitations was that, remarkably, the investigation did not support inefficient departments in improving their teaching and learning performance.

Lee and Johnes (2022) observed that the higher education literature focuses on the efficiency of the production of graduates and their employment, whereas more efforts should be made to allocate resources to achieve quality production. The authors applied a network DEA model to investigate 111 higher education institutions in England. The results revealed that not having a teaching qualification negatively impacts efficiency. Teaching qualifications allow instructors to reflect on their teaching approach and consider alternative methods to meet students' needs. However, a teaching qualification is not a teaching quality. A teaching qualification is based on standards consistent to all English institutions, while teaching quality is based on subjective evaluation. Evaluations imply the existence of a possible bias caused by the subjectivity of instructors (Lee and Johnes, 2022) and students (Valencia, 2022).

Investigating the relative performance of classes takes us one step further toward the micro perspective. Montoneri et al. (2012) applied a DEA model to explore the relative efficiency of 18 classes of first-year students following an English subject at a Taiwanese university. The authors identified key learning performance indicators and designed a mechanism for instructor performance improvement to help instructors focus their efforts on the didactic suggestions made and to support decision-makers in developing educational strategies.

The results by Montoneri et al. (2012) indicated that the richness of the subject content, followed by the level of positive attitude, were the most important factors and should be prioritised. The authors suggested that instructors should find a balance between motivating students and correcting their mistakes in class. In contrast, they suggested that students make efforts during learning and accept criticism, and that the attitude during the learning process could be influenced by cultural factors.

Considering classes is essential when investigating efficiency because the results can be directly translated into decisions at the operational level. However, this perspective is commonly less investigated than the macro level, probably due to the need for more data. To the best of the authors' knowledge, no investigation has proposed a DEA approach to investigate the efficiency of active-learning classes, or a sample encompassing passive- and active-learning classes.

2.3. Higher education in Chile

According to the World Bank (2022), Chile has the highest gross enrolment rates in Latin America. The Chilean higher education system is composed mainly of traditional universities, technical higher educational institutions, and professional institutes, with private colleges and private universities making up a significant share of the sector (Martin and Furiv, 2021).

The Chilean Ministry of Education oversees and regulates higher education. The ministry is responsible for supervising the quality of education, accrediting new institutions, and allocating state subsidies to the sector. Chile has implemented various policies and programs to promote quality and access to higher education, such as the programme *Mejoramiento de la calidad y la equidad de la educación superior* (MECESUP - Improvement of the quality and equity in higher education) to improve academic quality (Salazar et al., 2022). Additionally, a national monetary subsidy program exists for students who need financial assistance to pursue higher education. One issue facing the Chilean higher education sector is the tension between access and inclusion in a segmented university system (Jarpa-Arriagada and Rodríguez-Garcós, 2021).

An initial study evidenced that the previous gratuity and financial aid policies designed to increase access to private universities for students from families with lower levels of income and education had a marginal impact in Chile (Espinoza et al., 2022). Similarly, despite national policies in place in Argentina and Peru, students who need support tend to enrol in low-quality, non-selective private institutions (Mendoza, 2020). Therefore, while there has been progress in increasing

access to higher education, disparities persist in the quality of education and employment opportunities among students from different backgrounds (Mendoza, 2020).

Recently, Chile has considered a radically new constitution that enshrines bold commitments to higher education reform, including free higher education. However, much debate remains about whether such reforms are feasible and sustainable. Overall, the Chilean higher education sector is a complex, constantly evolving system subject to ongoing efforts to improve both access to higher education and its quality (Martin and Furiv, 2021).

According to Véliz et al. (2023), the literature on creating research capacity at higher education institutions in developing countries is recent and scarce. Learning and teaching may be the first steps toward a high-quality research capacity. Studies have been published regarding the efficiency of high education institutions in the Chilean context (Barahona-Urbina and Barahona-Droguett, 2023; Cossani et al., 2022; Ramírez and Alfaro, 2013). However, these studies focused on comparing Chilean universities rather than on courses at a particular university; they did not consider attitude during learning, and were not focused on teaching and learning performance. Therefore, our investigation is justified since it focuses on classes and students to improve the learning process, which is essential to adequate teaching practices in engineering courses.

In Chile, Cossani et al. (2022) proposed a new network DEA model for measuring universities' relative efficiency, jointly considering their research and teaching performance. The authors developed a model based on the slack-based measure (SBM) DEA model. Among the advantages of the SBM model is that it can be run without choosing an orientation, which is hard to justify in practical cases (Dyckhoff and Souren, 2022). The authors applied their model to 33 private and public Chilean universities. They concluded that the Catholic University of the North (UCN) ranked between 26th and 29th (depending on the model configuration).

In other words, the UCN is among the worst performers. The UCN is a private Chilean institution of higher education belonging to the Catholic Church. The UCN comprises seven faculties, 20 departments, eight schools, three institutes, seven centres dedicated to research, and a science and technology park. The main UCN campus is the Antofagasta campus, located in the northern economic zone. The engineering school that we investigated is located on the Coquimbo campus, which borders the central economic zone. Chile has 16 regions and six economic zones (Banco Central de Chile, 2023). In 2022, the Northern Region contributed 9.69 % of GDP, while the Central Region contributed 13.76 %. In contrast, the Metropolitan Region contributed 44.32 % to GDP, evidencing that the country's economy is unequal (Banco Central de Chile, 2023).

Therefore, our study focuses on a region that could usefully be targeted by policies to reduce the country's inequalities. Improving STEM teaching and learning processes has already been pointed out as a tool for economic development and capable of reducing economic inequalities among countries (De La Hoz et al., 2021). The same effect is expected to take place among regions within a country.

3. Materials and method

3.1. Design science research methodology (DSRM)

This section presents information about the empirical methodological procedures that feature in our article. In terms of the Design Science Research Methodology (DSRM), based on Peffers et al. (2007), Charles et al. (2019), and Tsolas et al. (2020), the activities of this study are shown in Table 1.

3.2. Data collection

The Engineering School of Coquimbo (EIC) is located on the UCN

Table 1
DSRM applied to the current study.

DSRM Activities	Description	Knowledge base
Problem identification and motivation	There is a need to improve teaching and learning processes to achieve quality education, focusing on different classes that adopt active and passive pedagogical techniques to ensure equal access and eliminate gender disparities in education. As a result, this need is related to addressing progress made towards attaining SDG 4, quality education for all, and would also help to solve the problems associated with SDGs 8 and 10, to ensure equal opportunity, decent work, and reduce inequality through quality education, addressing the progress towards achieving these SDGs, crucial for a country's growth and sustainable development.	Literature and real-world problems.
Define the objectives of a solution	This paper proposes a simple approach for assessing the relative efficiency of engineering classes. This approach must enable the joint measurement of classes that adopt active and passive pedagogical techniques. The slack-based measure (SBM) DEA model under the benefit of doubt (BoD) condition (with unitary input) was chosen because it is non-radial, thus, there is no need to assume proportional changes among measures, and all measures are "the higher, the better" when composing the composite index. Thus, it is possible to support educational decision-makers in identifying strengths and weaknesses in the classes of their institutions.	Literature and knowledge of existing tools: DEA.
Design and development	The methodological design of this paper is based on DEA, an approach applying the slack-based measure (SBM) DEA model under the benefit of doubt (BoD) condition (with unitary input) because it is non-radial. Thus, there is no need to assume proportional changes among measures, and all measures are "the higher, the better" when building the composite index. Also, besides the lack of studies evaluating classes instead of universities, the SBM under BoD condition is less applied to aspects related to the learning performance of students and teaching in the literature than common models in order to evaluate the progress made towards attaining SDG4 on "quality	DEA Model, SBM-VRS, BoD condition.

(*continued on next page*)

Table 1 (*continued*)

DSRM Activities	Description	Knowledge base
	education", SDG 8 on "decent work and economic growth", and SDG 10 on "reducing inequality within and among countries". Therefore, it is possible to support educational decision-makers to identify strengths and weaknesses in classes at their institution without using data from other institutions. This approach was applied to 70 classes comprising 38 engineering courses, including 11 that relied on active learning pedagogical approaches.	
Demonstration	Empirical analysis (case study demonstration using data from a Chilean university). The proposed approach is applied to evaluate engineering classes to assess a Chilean university's progress towards the SDG4 on "quality education". The measures are related to the quality of education and equality, the students and learning process, and the aggregated evaluations of the classes. It emerges that the most inefficient classes were the expositive ones, mainly in the first year of the course.	Applying the proposed approach using real-world data.
Evaluation	Evaluation of the benchmarks and comparative analysis to identify the efficient classes, as well as strengths and weaknesses of the classes. Classes leading up to graduation tend to be more efficient than those at the start of courses. In addition, active classes are more efficient than expositive ones. The results emphasise the significant impact of instructors' qualifications and the need for adopting strategies to standardise the efficiency level across all classes within the same course. Decision-makers can use this method to evaluate their institutions, adapt the model and measures to their reality, improve teaching and learning processes to achieve quality education, ensure equal access, eliminate gender disparities in education, and address the progress made towards attaining SDG 4, quality education for all. In addition, it would help them to solve problems related to SDG10, contributing to sustainable development and more decent work for all (SDG 8).	Understanding of current solution.
Communication	The findings suggest that improvements are needed in learning and teaching processes, including enhancing the qualifications of teaching staff, reviewing class size and assessment strategies towards equality, more successfully integrating digital technologies, expanding active-learning classes, and strengthening classes on fundamental subjects from the beginning of the engineering courses.	Communicating research findings of the application of the DEA SBM-VRS-BoD to improve Engineering Classes toward the achievement of SDGs 4, 8 and 10.

Note that the methodology procedures were organised in the following stages:
a) Data collection. Information about the data and the process. Further details are provided in the following subsection.
b) DEA SBM model under the benefit of doubt (BoD) condition. Configuration of the model and measures are presented in the following subsections.

campus of Coquimbo. The EIC offers three graduate engineering courses: Civil Engineering in Computing and Informatics (ICCI), Civil Industrial Engineering (ICI), and Information Technology Engineering (ITI).

The university adopts a comprehensive data collection methodology, applying a closed structured questionnaire. This survey is carried out online using the Banner enterprise resource planning (ERP) software system employed at the university, which the students answer before the end of their current academic semester. This information is obtained anonymously, aggregated, and sent to a central unit at the university to be distributed to each academic department. The survey questions are defined in the university's institutional educational project document (UCN, 2017). This project describes the expected characteristics of good teaching for the university, validated by the specialists of the university's institutional educational project. The data concerned the first semester of 2022, which can be considered a limitation of this study. Thus, we suggest collecting data from more periods.

The first semester of 2022 offered 38 subjects specific to engineering courses. These subjects are divided into several classes (named "paralelos" in Spanish) to keep the number of students per classroom small—ideally, less than 30 students per classroom. The instructors of classes on the same subject are not necessarily the same person. However, all instructors of a particular subject must follow the same syllabus.

Here, each class ("paralelo") was considered a decision-making unit (DMU) in the model. Thus, 70 DMUs were investigated, considering the data (information about the measures can be found in Table 2) from the EIC's educational system database for the first semester of 2022. Of these DMUs, 11 are mainly based on active learning pedagogical methodologies—problem-based learning (PBL). These DMUs have the word "project" in the title (Table 3).

In our study, all measures were of the type "the higher, the better," i.e., using all data in the same orientation as outputs, and only a unitary input, due to the BoD condition for creating a composite index without assuming the production perspective. Consequently, there is no need to explain why the input can "produce" the outputs in the adopted approach using the SBM-VRS-BoD model once the inputs are unitary, assuming equal levels among DMUs.

It is essential to apply empirical rigour and test the reliability and sample size of the questionnaire and data. The sum of individual student responses for the final dataset of the 70 engineering classes was 2021 responses from a total of 2368 students in the first semester of 2022. The reliability assessment by Cronbach's Alpha (equivalent to the average measures consistency intraclass correlation coefficient (ICC)) used to measure the internal consistency of a set of data resulted in "good", "relatively high", "fairly high", or "high" reliability (0.76 with a 95 % confidence interval between 0.70 and 0.81), depending on the reference, as it can be seen in Taber (2018).

Table 2
Detailed information about measures collected in the educational system database.

Short form	Measure	Description
GRADES	Aggregated average grades of students	Aggregated average grade per number of students
FEMALES	Percentage of women	% (Number of women per number of students)
APPROVED	Percentage of approved students	% (Number of approved students per number of students)
PEDAG	Build pedagogical strategies	% of students answered "yes" to "almost always" or "always" in that topic
TLP	Implement the teaching-learning process	% of students answered "yes" to "almost always" or "always" in that topic
SLA	Design of strategy for learning achievements	% of students answered "yes" to "almost always" or "always" in that topic
RESPECT	Respect for the dignity of people and inclusion	% of students answered "yes" to "almost always" or "always" in that topic
CONTEXT	Capitalise on experience and contextualisation of teaching	% of students answered "yes" to "almost always" or "always" in that topic
IICT	Integrating ICT in Learning Environments	% of students answered "yes" to "almost always" or "always" in that topic
ADM	Administrative aspects	% of students answered "yes" to "almost always" or "always" in that topic

3.3. DEA SBM model under the benefit of doubt (BoD) condition

This section presents the formulations of the slack-based measure (SBM) model from Tone (2001) under the benefit of doubt (BoD) condition (with unitary input). We also present the definition of the measures based on the choice of measures by Montoneri et al. (2012).

Each measure is described in Table 2. Ten outputs and one unitary input characterise the DMUs. These measures were chosen based on data recommendations from Montoneri et al. (2012). Among the outputs, seven measures are related to aggregated class evaluations. For each question, students can choose to answer "yes" to one of five possibilities ("never", "almost never", "sometimes", "almost always", or "always"); then the frequency of students that answered "almost always" and "always" are computed for each question as a sign of best practices. In this study, to evaluate the class, we used the percentage of students who answered "yes" to "almost always" or "always" on the topic in question. In other words, we used the percentage of the frequency of best evaluations. They are:

1. Build pedagogical strategies (PEDAG)
2. Implement the teaching-learning process (TLP)
3. Design of strategy of learning achievements (SLA)
4. Respect for the dignity of people and inclusion (RESPECT)
5. Capitalise on experience and contextualisation of teaching (CONTEXT)
6. Integrating ICT in learning environments (IICT)
7. Administrative aspects (ADM)

   The other three measures are related to the students and the learning process. They are:

8. Aggregated average grades of students (GRADES)
9. Percentage of approved students (APPROVED), and
10. Percentage of women (FEMALES)

According to Hollingsworth and Smith (2010), applying DEA under a Variable Returns to Scale (VRS) assumption is recommended when working with ratio data. Cook et al. (2014) clarify that the VRS assumption preserves the range of percentage projections between 0 %

Table 3
Efficiencies and rankings of the DMUs.

DMU	Semester	Efficiency	Ranking
Asset Management (Class 1)	9	1	1
Asset Management (Class 2)	9	1	1
Digital Organisation (Class 2)	9	1	1
Innovation Management (Class 2)	9	1	1
Project for the Formulation and Evaluation of Investment Projects (Class 1)	9	1	1
International Commerce (Class 1)	8	1	1
Operations Management II (Class 1)	8	1	1
SAP Integrated Processes (Class 1)	8	1	1
Strategic Planning (Class 1)	8	1	1
Supply Chain Design Project (Class 1)	8	1	1
Sustainable Development and Engineering (Class 1)	8	1	1
Electrotechnical (Class 3)	7	1	1
Human Capital Management (Class 1)	7	1	1
Economic Engineering (Class 1)	6	1	1
Operations Management I (Class 2)	5	1	1
Optimisation and Simulation Project (Class 1)	3	1	1
Optimisation and Simulation Project (Class 2)	3	1	1
Engineering Introduction Project (Class 2)	1	1	1
Electrotechnical (Class 2)	7	0.9727	19
Quality and Reliability Project (Class 1)	4	0.9602	20
Supply Chain Management (Class 2)	7	0.9484	21
Quality (Class 2)	3	0.9445	22
Quality (Class 1)	3	0.9422	23
Digital Organisation (Class 1)	9	0.9411	24
Engineering Introduction Project (Class 1)	1	0.9267	25
Mining Supply (Class 1)	8	0.9111	26
Sustainable Development and Engineering (Class 2)	8	0.9061	27
Supply Chain Management (Class 1)	7	0.8938	28
Engineering Drawing (Class 2)	5	0.8867	29
Engineering Business (Class 1)	8	0.8814	30
Engineering Drawing (Class 1)	5	0.8741	31
Human Capital Management (Class 2)	7	0.8724	32
Engineering Introduction Project (Class 3)	1	0.8612	33
Reliability (Class 1)	5	0.8602	34
Innovation Management (Class 1)	9	0.8583	35
Fluid Mechanics (Class 2)	7	0.8582	36
Finance (Class 2)	10	0.8544	37
Operations Research I (Class 2)	5	0.8521	38
Reliability (Class 2)	5	0.8436	39
Management Accounting (Class 1)	6	0.8295	40
Operations Research II (Class 1)	6	0.813	41
Productive Systems Design Project (Class 1)	6	0.8073	42
Finance (Class 1)	10	0.7953	43
Operations Management I (Class 1)	5	0.7931	44
Simulation (Class 2)	7	0.7898	45
Engineering Introduction Project (Class 4)	1	0.7673	46
Fluid Mechanics (Class 1)	7	0.7589	47
Operations Research II (Class 2)	6	0.7396	48
Engineering Introduction Project (Class 5)	1	0.7325	49
Data Science II (Class 1)	8	0.7299	50
Information Technologies (Class 1)	3	0.6842	51
Simulation (Class 1)	7	0.6575	52
Mining Processes (Class 1)	8	0.6547	53
Economy (Class 1)	4	0.654	54
Applied Statistics (Class 2)	5	0.6465	55
Economic Engineering (Class 2)	6	0.6383	56
Statistics (Class 3)	3	0.6353	57
Statistics (Class 1)	3	0.5745	58
Programming (Class 4)	2	0.5658	59
Applied Statistics (Class 1)	5	0.5495	60
Information Technologies (Class 2)	3	0.5404	61
Operations Research I (Class 1)	5	0.4908	62
Electrotechnical (Class 1)	7	0.4866	63
Statistics (Class 2)	3	0.4732	64
Programming (Class 1)	2	0.4692	65
Programming (Class 2)	2	0.4082	66
Programming (Class 3)	2	0.4032	67
Thermodynamics (Class 3)	6	0.3233	68
Thermodynamics (Class 2)	6	0.3097	69
Thermodynamics (Class 1)	6	0.2612	70

and 100 %.

We applied the SBM model under the BoD condition, i.e. unitary inputs $x_{ik} = 1$ $(i = m = 1;\ k = 1, \ldots, z)$, where z is the number of DMUs, and considering the outputs expressed as $\{y_{ik} \in \mathrm{R}_+\}$ $(j = 1, \ldots, n;\ k = 1, \ldots, z)$, where y_{jk} is the *j-th* output of the *k-th* DMU, and k=0 refers to the DMU under analysis. The SBM model under the VRS assumption is as follows in Eqs. 1 to 6 (Tone, 2001):

$$\text{minimise } \tau_0 = \mathrm{h} - \left(\frac{1}{\mathrm{m}}\right) \sum_{\mathrm{i}=1}^{\mathrm{m}} \frac{\mathrm{S}_\mathrm{i}^-}{\mathrm{x}_{\mathrm{i}0}} \tag{1}$$

subject to

$$\mathrm{h} + \left(\frac{1}{\mathrm{n}}\right) \sum_{\mathrm{j}=1}^{\mathrm{n}} \frac{\mathrm{S}_\mathrm{j}^+}{\mathrm{y}_{\mathrm{j}0}} = 1 \tag{2}$$

$$\sum_{\mathrm{k}=1}^{\mathrm{z}} \Lambda_\mathrm{k} \mathrm{x}_{\mathrm{ik}} + \mathrm{S}_\mathrm{i}^- - \mathrm{h}\mathrm{x}_{\mathrm{i}0} = 0 (i = 1, \ldots, m) \tag{3}$$

$$\sum_{\mathrm{k}=1}^{\mathrm{z}} \Lambda_\mathrm{k} \mathrm{y}_{\mathrm{jk}} - \mathrm{S}_\mathrm{j}^+ - \mathrm{h}\mathrm{y}_{\mathrm{j}0} = 0 (j = 1, \ldots, n) \tag{4}$$

$$\sum_{\mathrm{k}=1}^{\mathrm{z}} \Lambda_\mathrm{k} = \mathrm{h} \tag{5}$$

$$\Lambda_\mathrm{k} \geq 0\ (\forall \mathrm{k}),\ \mathrm{S}_\mathrm{i}^- \geq 0\ (\forall \mathrm{i}),\ \mathrm{S}_\mathrm{j}^+ \geq 0\ (\forall \mathrm{j}),\ \text{and } \mathrm{h} > 0 \tag{6}$$

where τ_0 is the efficiency of the DMU under analysis, h is the model linearization measure, S_i^- is the slack of the *i-th* input, S_j^+ is the slack of the *j-th* output, Λ_k is the contribution (intensity measure) of the *k-th* DMU to the DMU under analysis, Eq. 5 is the constraint representing the VRS assumption, and the other parameters were presented in the paragraph before presenting Eq. 1.

The optimum solution $(\tau_0^*, h^*, \Lambda_k^*, S_i^{-*}, S_j^{+*})$ is described by the conditions (Eq. 7):

$$\tau_{\text{optimal}} = \tau_0^*,\ \lambda_\mathrm{k}^* = \frac{\Lambda_\mathrm{k}^*}{\mathrm{h}^*},\ \mathrm{s}_\mathrm{i}^{-*} = \frac{\mathrm{S}_\mathrm{i}^{-*}}{\mathrm{h}^*},\ \mathrm{s}_\mathrm{j}^{+*} = \frac{\mathrm{S}_\mathrm{j}^{+*}}{\mathrm{h}^*}, \tag{7}$$

where λ_k^*, S_i^{-*}, and S_j^{+*} are the original optimal measures (before linearising) solutions, and the DMU under analysis is deemed efficient when $\tau_0^* = 1$.

4. Results

Table 3 shows the DEA results; 18 of the 70 classes were identified as efficient (25.7 %). Table 4 shows the number of efficient and inefficient classes that use active and passive pedagogical methodologies. 45.45 % of the active classes were efficient, compared to 22.03 % of the passive classes.

Every class from the second and fourth semesters was inefficient. This result can be partially explained by the fact that data were collected in the first semester. New students are accepted exclusively in the first semester. If students follow the regular planning, they will conclude the engineering graduation in the tenth semester. However, not all students are capable of following the regular planning. Students may be forced to interrupt their studies for several reasons, and some of them may also be reproved for a subject and forced to redo it. Thus, the school offers almost all subjects in all semesters, depending on the students' demands. For example, students doing in the first semester a subject planned to be done in the second semester are often students previously reproved in this subject or students who had to interrupt their studies for a while. Thus, classes may be inefficient in these semesters because most students in the course struggle more with these subjects. Cross-temporal studies are recommended to overcome our study's time constraint.

Table 4
Status (efficient or inefficient) and pedagogical methodologies (active or passive).

	Active	Passive
Efficient	5	13
Inefficient	6	46

Looking at the classes in the first and second years of engineering courses, only one efficient class was from the first semester. This class comes under the subject "Engineering Introduction Project (Class 2)", which is a subject that adopts active learning methodologies and PBL oriented to engineering projects. For this subject, only "Class 2" was efficient. Classes 1, 3, 4, and 5 occupied between 25th and 49th place. It is interesting to highlight that no class comes below the 53rd position, i.e., none is in the least efficient quartile. Two efficient classes were from the first semester of the second-year engineering course and concerned the same subject, "Optimisation and Simulation Project (Classes 1 and 2)". Since this subject was divided into only two classes, the whole subject was considered efficient. This subject combines expositive classes with active-learning methodologies based on PBL.

In the third year, regular students are expected to conclude basic engineering subjects and enter more professional fields. In addition, students are expected to start apprenticeships and internships during this period. Only one class was considered efficient in each semester, respectively, "Economic Engineering (Class 1)" and "Operations Management I (Class 2)", both of which were passive classes. "Economic Engineering" was divided into two classes, and Class 2 occupied 56th position – i.e., the least-efficient quartile. "Operations Management I" was divided into two classes, and Class 1 occupied 44th position, i.e., the third efficiency quartile.

In the fourth year, i.e., in the seventh semester, there were two efficient classes, "Electrical Engineering (Class 3)" and "Human Capital Management (Class 1)", both of which were passive classes. "Electrical engineering" was divided into three classes. Classes 2 and 1 occupied the 19th and 63rd positions, respectively. Like "Economic Engineering", "Electrical Engineering" had classes positioned in the most and least efficient quartiles. "Human Capital Management" is divided into two classes, and Class 2 occupies 32nd position.

In the tenth semester, two classes, "Finance (Class 1 and 2)", occupied the 43rd and 37th positions, respectively. Six classes were efficient in the eighth semester (fourth year) and five in the ninth semester (fifth year). This corresponds to 31.6 % and 26.3 % of the efficient classes. An observation that can be drawn is that the subjects taught in the last graduation semesters are more efficient.

In the eighth semester (fourth year), the efficient classes were "International Commerce (Class 1)", "Operations Management II (Class 1)", "SAP Integrated Processes (Class 1)", "Strategic Planning (Class 1)", "Supply Chain Design Project (Class 1)", and "Sustainable Development and Engineering (Class 1)", all of which concern different subjects. All classes are passive except for "Supply Chain Design Project" based on PBL. "International Commerce", "SAP Integrated Processes", "Strategic Planning", and "Sustainable Development and Engineering (Class 1)" only had one class. At the same time, the "Supply Chain Design Project" was divided into two classes, and Class 2 occupied the 27th position – i.e., the second efficiency quartile.

In the ninth semester (fifth year), the efficient classes were "Asset Management (Classes 1 and 2)", "Digital Organisation (Class 2)", "Innovation Management (Class 2)", and "Project for the Formulation and Evaluation of Investment Projects (Class 1)". The latter subject involved a single active class, while all the others were passive classes. Like the "Optimisation and Simulation Project" in the third semester (second year), the two "Asset Management" classes were efficient. Class 1 on "Digital Organisation" occupied the 24th position, and Class 1 on "Innovation Management" ranked 35th. The discrepancy among passive classes in a subject in the last semester was lower than in the first

semester.

Fig. 1 shows the distribution of the analysed classes among the engineering course semesters. The distribution of the efficient classes is shown in orange, and the most inefficient classes in grey. Efficient classes tend to take place towards graduation, while the most inefficient classes are concentrated in the second, third, fifth, and sixth semesters.

The least efficient cases feature exclusively passive classes (between 53rd and 70th positions). Most subjects in the least-efficient quartile had all their classes in the same quartile. These were: "Programming" (Classes 1–4, in the second semester), "Statistics" (Classes 1–3, in the third semester), "Applied Statistics" (Classes 1 and 2, in the fifth semester), and "Thermodynamics" (Classes 1–3, in the sixth semester). The exceptions were "Information Technologies (Class 2)" (third semester), "Operations Research I (Class 1)" (fifth semester), "Economic Engineering (Class 2)" (sixth semester), and "Electrical Engineering (Class 1)" (seventh semester). Class 1 of "Information Technologies" and Class 2 of "Operations Research I" occupied the 51st and 38th positions, respectively. Classes on "Economic Engineering" and "Electrical Engineering" have already been discussed among the efficient cases. In addition, some subjects had a single class in the last quartile. They were "Economy (Class 1)" (fourth semester) and "Mining Processes (Class 1)" (eighth semester).

No cases from the first and ninth semesters featured in the least-efficient quartile. Engineering graduation comprises nine semesters of classes (the tenth semester is reserved for internship). The inefficient classes were divided between the initial and final class semesters. Moreover, it is possible to state that the most inefficient classes were those related to fundamental engineering skills – i.e., statistics, programming, and thermodynamics – which require a high degree of mathematical thinking. This may imply that it is necessary to re-evaluate how these critical subjects are generally taught and consider new teaching and learning methodologies.

5. Discussions

According to the results of the analysis of the proposed approach applying SBM-VRS-BoD model, grades, respect (and inclusion), and ICT were the most important measures, and 45.45 % of the classes that adopt active learning methodologies and PBL oriented to engineering projects were efficient, compared to 22.03 % of the passive classes. Here we present a summary of the objective strategies.

Improving the implementation of Active Learning Methodologies: Increase the number of classes that use active learning methodologies, such as problem-based learning (PBL), which are more efficient and improve students' performance (grades).

Integrating ICT and Digital Technologies: Emphasize using ICT, digital tools, technology in the classroom, and other resources to enhance teaching and learning processes.

Improving Instructor Qualifications: we recommended policies that promote the adoption of active learning methodologies in engineering education, investment in instructors' training, particularly in effective and inclusive teaching strategies, and development programmes to facilitate changes, ultimately improving educational outcomes.

Reassessing Class Size and Evaluation Methods: Smaller class sizes and improved assessment techniques can lead to better student performance (grades) and higher class efficiency. Also, we recommend establishing continuous assessment and evaluation frameworks using the proposed index as a benchmarking tool, enabling ongoing monitoring of educational efficiency, and facilitating timely interventions.

Strengthening Foundational Subjects: Enhance support for fundamental engineering subjects like programming, statistics, and thermodynamics, often less efficient, to improve students' foundational skills.

In addition, we suggest that policies could foster innovation in pedagogical strategies through grants and funding opportunities for research in education technology and active learning methodologies.

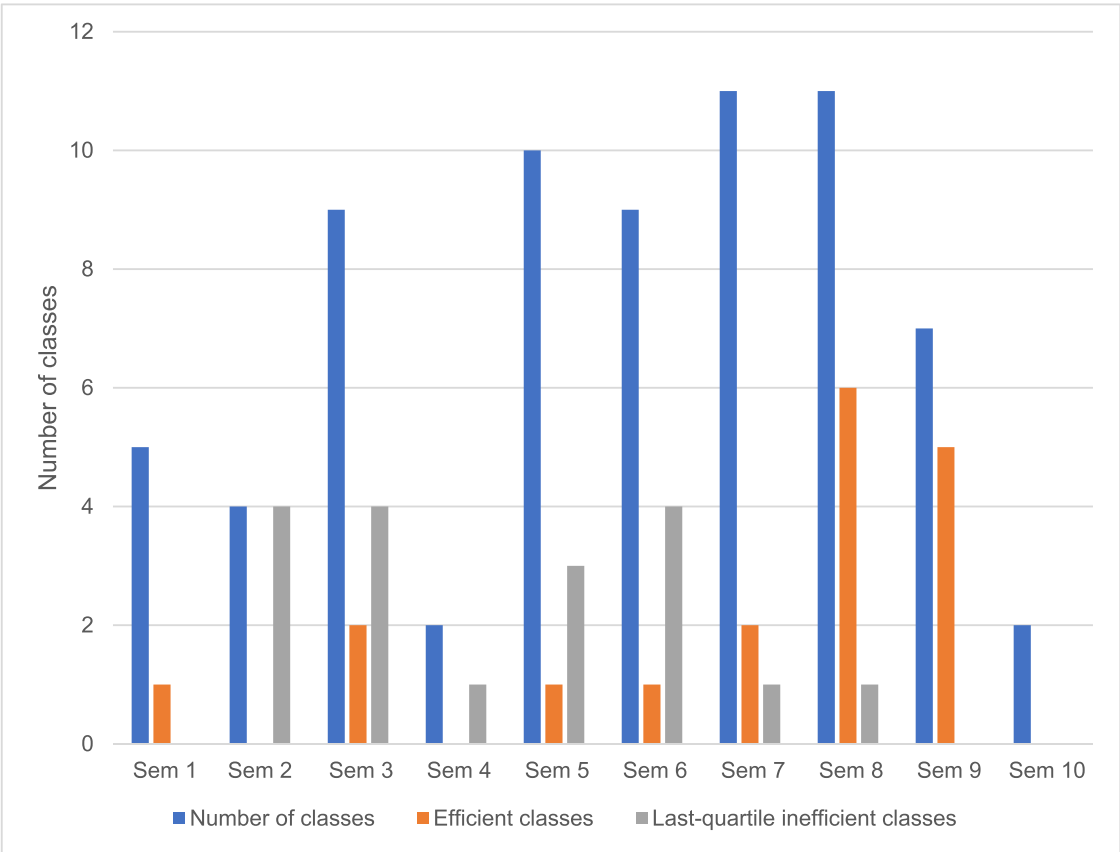

Fig. 1. Graph of the distribution of analysed, efficient, and most inefficient classes among engineering subjects.

We discuss these strategies, recommendations, and implications in the next subsections.

5.1. Learning efficiency and practical recommendations

Based on the results, three findings are evident: (i) the level of efficiency of classes is discrepant, even for classes of the same subject; (ii) classes related to fundamental engineering skills (mathematics and computation) tend to be inefficient; (iii) active-learning classes tend to be more efficient and less discrepant in terms of efficiency. Learning efficiency in passive-learning classes has been defined and measured in terms of closeness to ideal reference grades (Yang, 2024). On the other hand, learning efficiency in active-learning classes has been defined and measured in more complex ways, such as balancing students' opinions about workload and time spent developing activities, the final grades, and student-student interactions (Fidalgo-Blanco et al., 2018) or the integration between theory and students' capacity to solve problems in practice (Da Silva et al., 2018).

Here, we propose the definition of learning class relative efficiency as a composite index that measures strengths in the teaching-learning process toward inclusive quality education based on the instructor's assessment of students, students' assessment of the instructor, and inclusion criteria. Independently of the class pedagogical methodology, the instructor's assessment of students was measured by aggregated average grades of students and the percentage of approved students. Students' assessment of the instructor was measured by the instructor's integration of ICT in learning environments, building pedagogical strategies and selection of learning activities towards learning results, implementation of the teaching-learning process, design of the strategy for learning achievements, and capitalise on experience and contextualisation of teaching - e.g., use of real examples. Also, students assess the instructor's administrative aspects regarding the classes, as well as respect for the dignity of people and inclusion. The percentage of women was an inclusion criterion.

Fig. 2 and Table 5 summarise the average contribution of each measure to the relative efficiency results. Since there were ten measures in our proposed index, whether all contributed equally, we might expect a contribution of 10 % for each measure in a theoretical scenario. However, in this application, the measure "GRADES" (aggregated average grades of students) contributed around 20 %, showing that the assessment approaches greatly impacted the results. It is worth noting that passive and active classes may require different methods of assessment and approval. For example, in active classes, Schaefer and Copeland (2024) demonstrated that interactive mid-term performance feedback improves students' final grades. Similarly, the measure "APPROVED" (percentage of approved students) also contributed slightly above 10 % (Fig. 2).

Our proposed definition of learning efficiency evidences the importance of adequate assessments in both directions: the instructor's assessment of students and the students' assessment of the instructor. It is worth noting that students' grade orientation is not a synonym for learning orientation. A mutual relationship exists between constructing an effective assessment of learning and learning (Schuwirth and Van der Vleuten, 2011). Educational institutions should focus on promoting learning orientation instead of assessment orientation. Previous research indicates that conscientiousness, openness, and age significantly predict learning orientation, while neuroticism negatively predicts it (Tippin et al., 2012). Given this, it is recommended that activities on campus that enhance conscientiousness and openness, such as student associations and workshops, be promoted. Classes should also be held with students of different ages when possible. Thirdly, more psychological support should be available to better prevent and treat neuroticism.

Therefore, researchers and educational decision-makers must carefully plan and construct assessment techniques. A well-designed and validated assessment strategy should avoid subjectivity from students and the instructor. An example of subjective bias to be avoided is evidence demonstrating that instructors' gender affects students' assessments of the instructor (Valencia, 2022). Also, Fisher et al. (2024) showed that instructors tend to be worse evaluated in larger classes. Our school's passive classes tend to have more students than active classes, which may affect instructors' evaluations. Thus, besides redesigning assessment strategies, it is also recommended to adopt smaller class sizes to improve learning efficiency. Remarkably, once interactive feedback affects the efficiency of active learning, keeping the class size small or even reducing the class size is crucial to enable more personalized

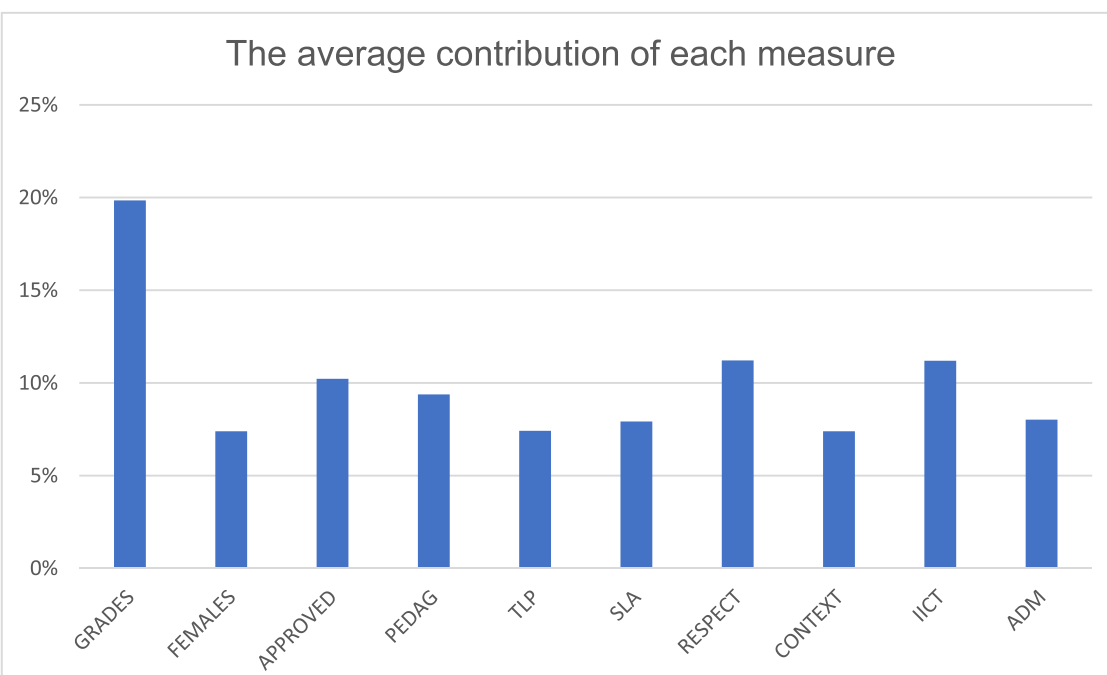

Fig. 2. Graph of the average contribution of each measure.

Table 5
The average contribution of each measure.

GRADES	FEMALES	APPROVED	PEDAG	TLP	SLA	RESPECT	CONTEXT	IICT	ADM
19.85 %	7.39 %	10.22 %	9.38 %	7.42 %	7.92 %	11.21 %	7.39 %	11.20 %	8.02 %

feedback.

The measures with the second and third largest contributions in Table 5 were "RESPECT" (Respect for the dignity of people and inclusion) and "IICT" (Integrating ICT in learning environments). Respect and inclusion are important, and Valdez and Kelp (2023) showed that inclusive practices like active learning are more perceived than explicit diversity, equity, and inclusion practices, suggesting the need for instructor training to support students regarding inclusion. Although the IICT is important, programming classes were all among the least efficient. This may imply that these technologies do not successfully improve students' computational and mathematical skills. This result demonstrates that the approach adopted here is valuable in helping educational decision-makers understand their strengths and weaknesses. We recommend that instructors receive careful training about integrating technology into their pedagogical strategies, focusing on improving students' computational and mathematical skills.

The measures with the lowest contributions in decreasing order were: PEDAG (Build pedagogical strategies), ADM (Administrative aspects), SLA (Design of strategy of learning achievements), and FEMALES (Percentage of women). The PEDAG measure can be related to active and passive learning since different pedagogical approaches require the construction of different strategies. For example, Da Silva et al. (2018) showed that bringing entrepreneurs/managers who present real problems to be analysed by the students in the PBL classes of a business management course can significantly improve learning and critical thinking. Thus, instructors in active classes should consider bringing different professionals to their classes. And critical thinking aligned with PBL improves learning efficiency and students' performance (Pu et al., 2019). Reducing the gender gap and increasing the number of women in STEM is a broad and necessary investigation theme. Evidence suggests that the gender of the teaching instructor and continuous interaction with a female role may improve women's performance and retention in STEM courses (Kapitanoff and Pandey, 2017). As already mentioned, a Chilean study demonstrated that instructors' gender affects students' assessments (Valencia, 2022). Further investigations are recommended on this theme.

Finally, the same teaching instructors are usually required to teach classes based on active and passive techniques at our school. Teaching uniform quality is a critical issue since evidence suggests that it impacts graduate employment outcomes (Lee and Johnes, 2022). Based on a network DEA model, Lee and Johnes (2022) suggested that the teaching staff could explain the relatively greater efficiency of some English higher education institutions. Similarly, given the current results, we recommend adopting strategies for leveraging teaching qualifications. As discussed, active classes require specific assessment techniques and pedagogical strategies. Also, proper and constant feedback plays a pivotal role in active approaches. Thus, we recommend having different and well-qualified instructors for each pedagogical approach and avoiding having the same staff to teach using both methodologies (i.e., active and passive). Instructors should also receive training on how to adapt their strategies for different class sizes (Wright et al., 2019). Remarkably, active learning instructors should be specifically trained to provide adequate feedback (Schaefer and Copeland, 2024).

5.2. Sustainable development goals (SDGs) and policy implications

Our research primarily focuses on educational efficiency and quality education (SDG 4 Quality education), yet its implications extend to other SDGs due to the interconnectedness of education and sustainable development. In the first place, our findings revealed that subjects taught in the last semesters before graduation are more efficient. Several studies have investigated students' adaptation difficulties in tertiary education, once students tend to drop graduation studies in the early years. Evidence suggests that higher education institutions can take active measures to minimise the negative impacts of adaption in the initial years and improve educational efficiency (Cielo et al., 2020; De Castro et al., 2018; De Medeiros Rosa et al., 2021; Nascimento and Beggiato, 2020; Pozobon and Lunardi, 2019). We recommend that policies should be drawn to improve student retention rates, remarkably in the first years.

A previous study in Chile demonstrated that educational policies were insufficient to support low-income students in overcoming social barriers and accessing higher education (Espinoza et al., 2022). Given the current educational system, students at our school are all of similar ages with a maximum of two years of variation. We recommend that policy-makers consider financing the education of students without age restrictions. We propose this based on three main reasons: (i) Low-income students face more financial, economic, and social difficulties, which reverberates in the need for more time to reach tertiary education when compared to high-income students; (ii) Age is pointed out as a main factor motivating students toward learning orientation (Tippin et al., 2012). Thus, mature students are expected to be more learning motivated and influence others, remarkably in active classes where the interaction student-student is promoted, evaluated, and associated with efficiency (Fidalgo-Blanco et al., 2018); (iii) Accepting older students enables mothers who had kids at a younger age to achieve professional qualifications and better incomes, enabling the inclusion of women in the labour market (SDG 5 Gender equality and SDG 8 Decent work and economic growth) and reducing poverty and inequalities (SDG 1 No poverty and SDG 10 Reducing inequalities).

We recommend policies that promote the adoption of active techniques in engineering education. Investment in instructors' training and development programmes can facilitate changes, ultimately improving educational outcomes. This involves establishing continuous assessment and evaluation frameworks using the proposed index as a benchmarking tool, enabling ongoing monitoring of educational efficiency, and facilitating timely interventions. In addition, we suggest that policies could foster innovation in pedagogical strategies through grants and funding opportunities for research in education technology and active learning methodologies.

For instance, our study contributes to achieving decent work and economic growth (SDG 8) by targeting STEM higher education, which plays a crucial role in narrowing economic disparities and fostering inclusive societies. Improving the knowledge and skills of students from diverse backgrounds from both genders also aids in reducing the knowledge gap between privileged and underprivileged communities, thereby ensuring inclusive and equitable education for all (SDG 4) and reducing inequalities (SDG 10). Given the pivotal role of quality education (SDG 4) in attaining other SDGs, aligning policy recommendations with the SDGs highlights the transformative power of education in broader societal and economic development. Policy-makers are urged to integrate these implications into their strategic planning and implementation processes to advance sustainable development goals efficiently.

5.3. Advantages of the proposed approach and model

When compared to previously published works, our paper considers several measures from teaching-learning process, e.g., the use of ICT or the inclusion aspect of quality education of SDG4, besides other

instructors' and students' assessment and performance indicators, with discussion about the contribution of each measure to the efficiency, and also comparing active learning methodologies to the traditional ones, resulting in more evidence about the efficiency of active learning methodologies.

Considering the pivotal role of educational efficiency and quality in sustainable development, our research offers substantial policy implications in line with several SDGs, mainly SDG 4 (Quality Education). Through pinpointing efficient pedagogical methods, the index proposed here offers policy-makers a strategic framework for more efficient resource allocation, ultimately enhancing educational access and quality.

The index proposed here is based on DEA, an approach applying the slack-based measure (SBM) DEA model under the Variable Returns to Scale (VRS) assumption and considering the benefit of doubt (BoD) condition (with unitary input). Our findings highlight the value of using the SBM-VRS-BoD model to evaluate the efficiency of engineering classes at a micro-perspective. By applying this model, we identified specific strengths and weaknesses within the classes, providing actionable insights for educational decision-makers. Our methodological choice has the advantage of not assuming proportional changes among measures, which is often hard to justify in practical applications. Therefore, it is an appropriate methodology to support educational decision-makers in identifying strengths and weaknesses in classes of any course at their institutions.

Additionally, all measures were of the type "the higher, the better," i.e., using all data in the same orientation as outputs and only a unitary input due to the BoD condition for creating a composite index without assuming the production perspective. Consequently, there is no need to explain why the input can "produce" the outputs in the adopted approach using the SBM-VRS-BoD model once the inputs are unitary, assuming equal levels among DMUs. This makes the index flexible and expands its applicability to any educational context, independent of the size of the institutions or pedagogical approaches.

6. Conclusions

This paper proposes a straightforward approach for assessing relative engineering class efficiency using a DEA-SBM-BoD model. It was applied to classes at a Chilean engineering school. This sample encompasses classes applying both active and passive learning techniques. The proposed approach is considered universal and can be employed by educational institutions to identify their strengths and weaknesses.

Regarding our application, the first conclusion of the results is that classes on the same subject have different levels of efficiency. Educational decision-makers are recommended to invest in ensuring adequate teaching training to standardise classes. Remarkably, the efficient dispersion among classes of the same subject is higher whether or not the subject is expositive (passive). This may be because the same instructors frequently teach classes based on active and passive techniques with different numbers of students per class. Educational decision-makers are recommended to rethink class sizes and provide teaching staff with adequate training regarding adapting strategies according to class size and teaching technique. In addition, it is worth noting that passive classes are the most inefficient. Decision-makers should consider expanding the number of classes based on active learning techniques and improving instructors' training in delivering expositive classes, especially for fundamental engineering subjects (e.g., programming, statistics, applied statistics, and thermodynamics).

A second conclusion drawn from the application is that classes taught towards the end of graduate studies tend to be more efficient than earlier classes. Previous literature centred on other Southern American countries evidences that students have more adaptation difficulties in the initial years of their studies and are more likely to drop out in these years (Cielo et al., 2020; De Castro et al., 2018; De Medeiros Rosa et al., 2021; Nascimento and Beggiato, 2020; Pozobon and Lunardi, 2019). Educational decision-makers are recommended to invest in measures that mitigate negative adaptation impacts in the first years of higher education. Further studies are recommended to extend the investigation of the theme in the Chilean context and a wider comparative Latin American context to understand which measures are likely to impact more classes' relative performance.

The third conclusion is that the proposed approach is heavily based on the results of students' assessment of teaching quality and teaching staff's assessment of students' learning - i.e., grades. Therefore, the assessment techniques are essential and must be constructed carefully. Remarkably, classes based on active and passive techniques apply different assessment methods.

This study pioneers a DEA (SBM-VRS-BoD) approach in this domain, introducing a novel efficiency index for jointly evaluating classes utilising active and passive pedagogical techniques, yet this novelty also poses a limitation. Further research is urged to devise assessment methods for comparing different pedagogical techniques. Additionally, exploring the role of digital transformation in teaching and learning while considering various pedagogical methods is advised. Also, the index relies on aggregated data from a single South American institution, underscoring data and sample size constraints. Expanding the sample and data collection across diverse educational settings globally is essential to fortify the index's robustness and applicability. Moreover, incorporating cross-temporal and cross-cultural analyses can offer insights into global educational efficiency. Addressing the limitations of static analysis, future studies should explore network, dynamic, or stochastic DEA models to discern the impact of teaching strategies on long-term student success. Lastly, while considering gender disparities in STEM courses, further investigations should explore strategies to narrow the gender gap in engineering education.

Thus, scholars and educational leaders stand poised to deepen their grasp of advancing educational practices for the betterment of students, institutions, and society. By confronting these constraints and delving into proposed avenues for future research, they pave the way for continual strides toward sustainable development, rooted in the principles of quality education (SDG 4), economic prosperity (SDG 8), and social equity (SDG 10).

CRediT authorship contribution statement

Paul Leger: Writing – review & editing, Writing – original draft, Validation, Resources, Funding acquisition, Data curation, Conceptualization. **Isotilia Costa Melo:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Conceptualization. **Paulo Nocera Alves Junior:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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